

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Fluctuations in supply/demand & energy storage - Set 1

Mark scheme

AQA spec 3.3.4.1; Energy resources; textbook pp. 217-224

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Lowest overnight (about 25 GW at 04:00), rising steeply from 06:00 to about 40 GW by 08:00 (1) • Fairly steady through the day then a peak of about 50 GW at 18:00 (evening) before falling overnight (1)	2
01.2	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • 24-hour day / night cycle: people asleep overnight; morning rise as heating, lighting, cooking, industry and offices start (up to 2) • Evening peak: lighting, cooking, heating as people return home; short-term 'TV pickup' at programme breaks / mealtimes (up to 2) • Other causes: seasonal (winter heating / lighting, summer air conditioning); weather-related (cold snaps); weekday / weekend differences in industrial use (up to 2)	4
01.3	<i>1 mark for each point.</i> • They generate 24 hours a day because it is uneconomic / slow to turn them off and on, and output changes slowly (1) • Demand falls faster at night than output can be reduced, so the surplus must be stored or is lost as heat from cables (1)	2
01.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Intermittent renewable resources: wind, solar, tidal (1) • Bulk delivery of transported fuels (oil tankers, coal trains, biofuel harvests) (1) • Breakdowns / maintenance of power stations (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 221-222; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	1 mark for each point. • At times of low demand / surplus supply (night, windy periods) surplus electricity pumps water from the lower to the upper reservoir, storing it as gravitational potential energy (1) • At peaks in demand water flows down through the turbines generating electricity (1) • Can go from standby to full power in under 15 seconds so it meets sudden peaks (TV pickup) that fossil / nuclear stations cannot follow (1)	3
02.2	A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown. • Mass = $7 \times 10^6 \times 1000 = 7 \times 10^9$ kg (1) • PE = $m g h = 7 \times 10^9 \times 9.8 \times 500$ (1) • = 3.43×10^{13} J (accept 3.4×10^{13}) (1)	3
02.3	1 mark for each point. • The electricity used for pumping is surplus that would otherwise be wasted / lost as heat, or cheap off-peak electricity (1) • It avoids building rapid-response low-efficiency peaking plants (open-cycle gas turbines) and allows intermittent renewables to be used (1)	2
02.4	Any 2 from the points below - 1 mark each, maximum 2. • Two water bodies / sites with a large height difference close together (mountainous) (1) • Impermeable, stable rock (1) • Close to the grid / demand (1) • Adequate water supply (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222; AQA spec 3.3.4.1.	
03.1	1 mark for each point. • Surplus electricity (eg windy nights) electrolyses water to produce hydrogen which is stored (1) • Hydrogen has a high energy density so it can be transported / stored for long periods (1) • Released by combustion for heat, industrial processes or steam turbines; or by fuel cells combining hydrogen and oxygen electrochemically to produce electricity and water (1) • Can also be used as a vehicle fuel or converted to methane / ammonia (1)	4
03.2	Any 3 from the points below - 1 mark each, maximum 3. • Compressed gas (up to 700 times normal volume) - energy to run the compressor, strong tanks (1) • Liquefied at very low temperature - energy for refrigeration and compression (1) • Adsorbed on a metal matrix (metal hydride) at low pressure - tanks larger and heavier than a petrol tank storing the same energy (1) • Conversion to ammonia - some energy needed but less than the alternatives (1)	3

Question	Marking guidance	Marks
03.3	<i>1 mark for each point.</i> - Both convert chemical energy to electricity electrochemically, but a battery stores a fixed amount of chemicals which are reformed by recharging (1) - A fuel cell keeps producing electricity as long as fresh fuel (hydrogen / alcohol) and oxidant are supplied (1)	2
03.4	<i>1 mark.</i> Using hydrogen made from surplus renewable electricity as the main store and carrier of energy, released when demand exceeds supply or intermittent sources fail, for heating, vehicles and electricity (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 219-221; AQA spec 3.3.4.1.	
04.1	<i>1 mark for each point.</i> - Vehicles: high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater; many cycles and high efficiency (1) - Stationary storage: mass is unimportant so the low energy density of lead-acid does not matter; lowest cost per kWh; well-established, recyclable (1) - Lead-acid drawbacks: fewer cycles and lower efficiency; lithium fire risk needs management (1) - Nickel-cadmium largely replaced because of cadmium toxicity and cost (1)	4
04.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Recharging speed (1) - Self-discharge during non-use (1) - Availability of raw materials (lithium, cobalt) (1) - Recyclability / disposal (1)	2
04.3	<i>1 mark for each point.</i> - Parked electric vehicles plugged into the grid form a very large distributed battery (1) - At peaks a small proportion of their stored energy is fed back to the grid; surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines (1)	2
04.4	<i>1 mark for each point.</i> - Surplus electricity electrolyses water to hydrogen, which can be converted to methane (1) - Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	
05.1	<i>1 mark for each point.</i> - Surplus energy drives a pump compressing air stored in underground caverns (salt mines); releasing it drives machinery / turbines (1) - Compression produces a lot of heat; storing this heat (hot oil / molten salt) and returning it when the air expands recovers energy that would otherwise be lost (1)	2

Question	Marking guidance	Marks
05.2	<i>1 mark for each point.</i> • As volume increases the surface area : volume ratio decreases (1) • Heat is lost through the surface so there is less surface per unit of stored heat; very large stores can carry summer solar heat into winter (inter-seasonal storage) (1)	2
05.3	<i>1 mark for each point.</i> • Potassium nitrate has a much higher boiling point so it stores heat at up to 550 °C without pressurisation (1) • Water could only reach such temperatures under high pressure, increasing costs; high temperature needed to raise steam for turbines (1)	2
05.4	<i>1 mark for each point.</i> • Materials with high specific heat capacity (concrete, water) warm and cool slowly (1) • They absorb heat in hot weather reducing overheating / air conditioning, and release it as the weather cools, delaying the need for heating (1)	2
05.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Flywheels: rotating mass stores kinetic energy for short-term smoothing / generation (1) • Supercapacitors: electrochemical capacitors with 10-100 times the energy density of conventional capacitors; possible future large-scale electricity storage (1) • Thermal storage in molten salt / high-volume tanks; hydrogen / P2G; batteries (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 222-224; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
06.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Renewables are intermittent (solar at night, calm days, slack tides) and some unpredictable; demand fluctuates daily, weekly, seasonally and short-term; supply and demand rarely match • Without storage, surplus is wasted and shortages must be met by fossil-fuel peaking plants; large-scale electricity storage not yet possible directly so energy must be converted • Pumped-storage HEP: proven, rapid response, peak shaving, but limited sites • Batteries: lithium-ion for grid and vehicles; factors affecting viability; V2G • Hydrogen: electrolysis, storage methods, fuel cells, hydrogen economy, P2G into the gas grid; high energy density enables vehicle use • Compressed air with heat recovery; thermal storage (molten salt for CSP, high-volume stores, thermal mass); flywheels; supercapacitors • Grid management / ICT and diversity of resources (wind + solar + tidal + biofuel) reduce but do not remove the need for storage; energy conservation reduces peaks • Evaluation of efficiency losses, costs, resource limits (lithium), and which technologies suit short-term vs inter-seasonal storage; judgement	9
06.2	<i>1 mark.</i> • Storing surplus energy at times of low demand / high supply to meet later peaks in demand, so peak generating capacity can be reduced (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 217-224; AQA spec 3.3.4.1.	